KAIZEN MANAGER AI

Product Document v1 — Complete Product Flow

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1. WHAT THIS PRODUCT IS

> A factory suggestion scheme where **AI does 90% of the work.**
> The operator spends **under 3 minutes.** AI writes the report, calculates the money, challenges the idea, predicts the obstacles, creates the project plan, and tells the Plant Head what to do about it on Friday morning.

What it replaces: the suggestion box, the paper Kaizen form, the Excel tracker, and the supervisor who spends 3 hours a week typing up other people's ideas.

The one rule everything is judged against: if an operator cannot submit an idea in under 3 minutes, they will stop using it by week 2. Every screen, every question, every field in this document exists only if it survives that test.

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2. THE PRODUCT IN ONE PICTURE

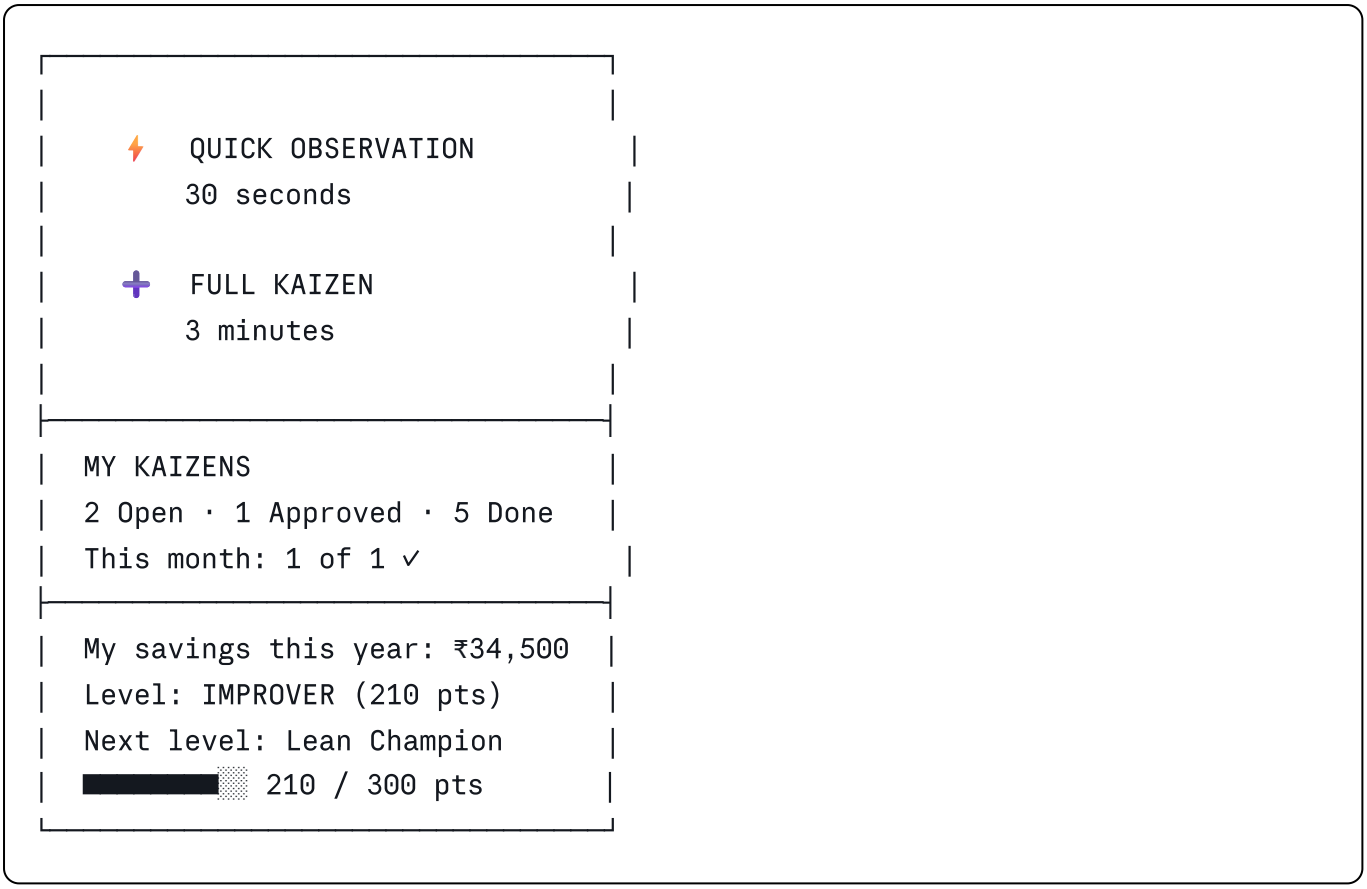
```
OPERATOR (3 minutes)              AI (instant)                MANAGEMENT
(zero effort)
_____________________             _____________
_________________________

Speak the problem          →    Transcribe + classify
Take a photo               →    SEE what's in the photo
Answer 4-6 questions       →    Write full Kaizen report
                           →    Calculate money + ROI
                           →    CHALLENGE the idea
                           →    Predict success + obstacles
                           →    Find similar past Kaizens
                                                    →   Business Impact
Card
                                                    →   One-tap approve
                                                    →   Auto-created PDCA
project
                                                    →   Friday Executive
Digest
                                                    →   Repeat-problem
alerts
```

PART A — THE OPERATOR JOURNEY

(This is the only part an operator ever sees. Total time budget: 3 minutes.)

SCREEN 0 — Home



Two ways in, deliberately. Most of what an operator notices is small — a loose guard, an unlabelled bin — and should take 30 seconds, not 3 minutes. **Quick Observation** is photo + voice only, no interview, no cost engine. **Full Kaizen** runs the whole journey below. AI promotes an observation to a full Kaizen if it turns out to be significant. Expect ~80% Quick, ~20% Full. *(Full rationale in Part F.)*

Everything else on the screen is secondary. No menus, no login-per-use — the phone stays logged in.

Steps 1–13 below describe the Full Kaizen track.

STEP 1 — Speak the Problem (30 seconds)

The operator taps the mic and simply talks. **No typing at any point.**

Example — Operator on Line 2:

> *"Every shift I walk about 20 metres to another machine to get the toolbox. Four of us share it."*

AI transcribes and cleans it up into a clear line:

> *"Operator walks ~20m each shift to a shared toolbox located at another machine. Shared by 4 operators."*

Why speaking, not typing: typing is the single biggest reason paper and app-based suggestion schemes die on the shop floor. Removing typing is not a convenience — it *is* the

adoption strategy. An operator who is uncomfortable writing will happily talk for 20 seconds.

STEP 2 — Take a Photo → AI SEES IT ⭐ *(Key differentiator)*

The operator takes one photo of the situation. That is all they do.

But the photo is not just stored as evidence — **Vision AI reads it.**

Example — operator photographs the toolbox at Machine 4. AI comes back with:

```
┌──────────────────────────────────────────┐
│  📷 I looked at your photo. I can see:   │
│                                          │
│  ✓ Toolbox on floor beside Machine 4     │
│  ✓ Tools loose inside — no shadow board  │
│  ✓ Walking path crosses forklift aisle   │
│                                          │
│  I also noticed 2 things you may not     │
│  have meant to report:                   │
│                                          │
│  ⚠ Possible oil patch near machine base  │
│  ⚠ Tools not labelled / not tagged       │
│                                          │
│  Do you want to include these?           │
│  [ Add to this Kaizen ]  [ Not relevant ] │
└──────────────────────────────────────────┘
```

Why this is the differentiator: the operator came to report *walking distance*. AI found a **5S gap, a safety hazard, and a possible leak** in the same photo — issues the operator had stopped noticing because they see them every day. One photo becomes three improvements.

⚠️ Three design rules that must be built in from day one

These decide whether this feature is loved or hated:

Rule 1 — AI never reports what the operator didn't approve. Vision findings are shown **only to the operator**, as a suggestion. If they tap "Not relevant," it is deleted and never reaches a supervisor.

Why: the moment operators believe the camera is reporting them to management, they stop taking photos. Then Step 2 dies, and with it the whole product. This must be a hard technical rule, not a policy note.

Rule 2 — AI always speaks in "possible," never in "confirmed." "Possible oil patch" — not "Oil leak detected." Vision AI *will* misread a wet floor as oil, a shadow as a crack, a rag as debris. Confident wrong statements destroy trust faster than no feature at all.

Rule 3 — Only 5 categories of finding are shown. 5S/housekeeping · Safety hazard · Leak or spill · Missing label/identification · Obstruction or access issue. Not "everything AI can see." A list of 15 observations makes the operator scroll past all of them.

STEP 3 — Confirm the Category (5 seconds)

AI has already classified it. Operator just confirms.

```
I think this is:  MOTION WASTE
[ Yes, correct ]   [ Change ]
```

Categories available if they change it: Motion · Waiting · Transport · Inventory · Quality · Safety · Energy · Productivity · Cost · 5S · Maintenance · Other.

Why AI classifies first: asking a shop-floor operator to pick the right waste category *before* explaining the problem is asking them to do abstract classification. That's exactly the form-filling this product exists to kill.

STEP 4 — The AI Interview (60–90 seconds)

Instead of a long form, AI asks **short questions, one at a time**, like a WhatsApp chat.

Example:

```
AI:  How many people use this toolbox?
     [ 1 ] [ 2-3 ] [ 4-6 ] [ 7+ ]        ← tap, don't type
👤   4-6

AI:  How many trips per shift, roughly?
👤   12

AI:  How long is one round trip?
     [ <30 sec ] [ 30-60 sec ] [ 1-2 min ] [ >2 min ]
👤   30-60 sec

AI:  Does this station ever make the line wait?
👤   No

AI:  What's your suggested fix?
👤   Buy another toolbox — around ₹12,000
```

Five questions. Done.

How the questions are chosen

Each category has its **own short question tree of 4–6 questions max.** There is no single universal question list.

Category	What AI asks about
Motion / Transport	Distance · frequency · people affected · cost of fix
Quality	Defect type · how often · scrap or rework cost · current containment
Safety	Near-miss or actual · how severe · how often exposed
Waiting	What is waited for · how long · how often · who waits
Energy	Equipment · running hours · idle time

And AI skips anything it already knows. If the operator said "20 metres" in Step 1, or the photo showed a missing shadow board, AI does not ask again. Repeating a question the operator already answered is the fastest way to feel like a government form.

⚠️ The bottleneck question — asked only when it applies

> *"Does this station ever make the line wait?"*

This is the **bottleneck question**. It decides whether a time saving is real money or wishful thinking (see Step 6). But it is **not** asked on every Kaizen — many ideas have nothing to do with a machine or a line at all. Asking a canteen or office Kaizen whether it makes the line wait makes the AI look stupid and wastes one of the five question slots.

AI asks it only when both are true:

1. The Kaizen is tied to a production station, machine or line, **and**
2. The claimed benefit is **saved time** — cycle time, changeover/setup, motion, waiting, transport or handling

AI skips it entirely when the saving is a direct cost, not time:

Kaizen type	Bottleneck question?	Why
Cycle time reduction	**Ask — most important case**	At the bottleneck this adds real output and is a big hard saving; anywhere else it adds nothing at all
Changeover / setup time (SMED)	**Ask**	Same logic — recovered machine hours only count if that machine limits the plant
Motion / waiting / transport at a machine	**Ask**	Time saved only becomes money if the line was waiting
Scrap / rework reduction	Skip	Material and rework cost is saved regardless of bottleneck
Energy / compressed air / lighting	Skip	kWh saved is cash saved, bottleneck is irrelevant
Consumable or tool life	Skip	Direct purchase cost avoided
Safety / near-miss elimination	Skip	Value is risk removed, not output gained
Office, stores, canteen, admin	Skip	No line to be a bottleneck of
5S / housekeeping (no time claim)	Skip	Benefit is order and safety, not throughput

⚠️ Cycle time savings use a different formula from motion savings

Do not calculate both the same way. This is the most common mistake in Kaizen savings claims:

	Motion / walking saved	**Cycle time saved at the bottleneck**
What is recovered	Operator hours	Machine/line hours → **extra sellable units**
Formula	hours saved × labour rate	extra units × **contribution margin per unit**
Typical size	Thousands of ₹	**Lakhs** of ₹
Counts as hard saving when	Time is redeployed or headcount changes	Almost always — extra output is sold

Example: cycle time on the bottleneck press cut from 48 sec to 42 sec.

```
6 sec saved × 3,000 cycles/shift × 3 shifts × 300 days = 1,500 machine-
hours/yr
→ ~4,50,000 extra parts/yr capacity
→ × ₹1.80 contribution per part = ₹8,10,000/yr HARD saving
```

Same 6 seconds on a **non-bottleneck** machine: **₹0 hard saving.** The machine idles 6 seconds longer per cycle and the plant ships exactly the same number of parts.

One extra check before crediting the full number: extra capacity is only money if there is demand to sell it into. If the plant is already producing to order and can't sell more, cycle time gains at the bottleneck convert to *cost avoidance* (overtime or a shift removed), not extra contribution. AI should ask the approver once: *"Can the plant sell the additional output, or does this remove overtime/a shift?"* — and calculate accordingly.

What this means overall: the bottleneck test applies to **all time-based savings** — cycle time, changeover and motion alike. Scrap, energy, consumables and tooling savings are **hard savings from the moment they're verified**, with no bottleneck condition attached. Don't apply the hard/soft split to every Kaizen; apply it where it belongs, or you'll under-report genuine cash savings and lose credibility in the other direction.

STEP 5 — AI Checks the Plant's Memory ⭐

Before writing anything, AI searches **every Kaizen ever submitted in this plant**.

Example — the operator sees:

```
🔍 Found 3 similar Kaizens in your plant:

- KZN-0142 — Line 1, March
  "Toolbox shared across machines"
  Status: Implemented ✓ Saved ₹31,000/yr
  → What they did: relocated toolbox to a
    trolley between both machines

- KZN-0207 — Assembly, May
  Status: Pending purchase approval

[ Mine is the same issue ]  [ Mine is different ]
```

Three things this achieves at once:

1. **Stops duplicates** — no more 6 people submitting the same idea and 5 getting rejected (which is how operators lose faith in the scheme).
2. **Spreads what already worked** — Line 1 solved this in March; Line 2 doesn't have to reinvent it.

3. **Feeds the repeat-problem detector** — see Part C.

STEP 6 — AI Calculates the Money (instant, no operator input)

Where the numbers come from — the Plant Cost Master

The operator is never asked for a rate, a cost or a margin. They only report what they can see: how many trips, how many seconds, how many parts, what the fix might cost.

Every rate used in the calculation is a **plant-level constant.** But they are **not all collected at once** — that would turn onboarding into a 30-field form nobody finishes.

GROUP A — Mandatory at enrollment *(6 fields, 10 minutes, blocks go-live)*

These are universal, non-sensitive, and every plant knows them off the top of their head. The system does not go live without them.

Master data	Example	Used for
Labour rate per hour (blended)	₹150/hr	Motion, waiting, handling savings
Working days per year	300	Every annualised figure
Shifts/day and hours/shift	3 × 8 hrs	Every annualised figure
Plant-average machine hour rate	₹850/hr	Cycle time, changeover, breakdown
Electricity rate	₹8.50/kWh	Energy Kaizens
Compressed air cost	₹1.20/m^3	Air leak Kaizens (very common, very cheap wins)

Ten minutes with the Plant Head and Accounts and you're live. No plant refuses these — none of them are commercially sensitive.

GROUP B — Captured on demand, from the approver ⭐

Everything else is asked **only when a Kaizen actually needs it** — and asked of the **approver**, not the operator, because a supervisor or manager knows the number and an operator doesn't.

Master data	Example	Asked when
Contribution margin per part	₹1.80/part	First cycle-time Kaizen on a bottleneck
Machine hour rate for a *specific* machine	₹1,240/hr for CNC-4	First Kaizen on that machine
Material rate by grade	₹78/kg MS · ₹310/kg SS	First scrap Kaizen using that material
Scrap sale recovery rate	₹22/kg	First scrap Kaizen
Overtime multiplier	1.5×	First capacity-release Kaizen
Is this station a bottleneck?	Press-2: Yes	First time-saving Kaizen on that station

How it appears — one line inside the approval card, never a separate form:

```
┌──────────────────────────────────────────────┐
│  KZN-0407 · Scrap reduction, Blanking        │
│                                              │
│  Saving: ₹1,42,000/yr    ⓘ PROVISIONAL       │
│                                              │
│   ❓ One number needed to finalise this:     │
│      MS sheet material rate? [  ₹___  /kg ]  │
│      (saved for all future Kaizens)          │
│                                              │
│  [ Approve ]  [ Reject ]                     │
└──────────────────────────────────────────────┘
```

⚠️ Scrap and rework cost is NOT one number per plant — it varies by product AND by stage

This is where a naive design breaks. A plant running 200 part numbers has 200 different scrap costs, and the *same part* has a different scrap cost depending on where it was rejected.

Two parts scrapped, same material, wildly different loss:

	Rejected at blanking	Rejected at final inspection
Material (2.4 kg MS @ ₹78)	₹187	₹187
Processing added so far	₹0	₹214 (blank + form + weld + machine)
Less scrap sale recovery (@ ₹22/kg)	−₹53	−₹53
Real loss per part	**₹134**	**₹348**

A system storing "scrap cost = ₹134/part" would under-report the second case by 60%.

The fix: store the small stable lists, calculate the rest

Store permanently *(a short list, changes rarely)*

- Material rate per grade — MS, SS304, aluminium, etc. Typically **5–15 entries** for an auto-ancillary plant, not 200.
- Scrap sale recovery rate ₹/kg — plants sell their scrap back; ignoring this overstates every scrap saving by 20–30%.

Calculate per Kaizen *(never stored per part)*

```
Scrap loss per part
= (part weight × material rate)
+ (cumulative processing cost up to the rejection stage)
− (part weight × scrap recovery rate)

where processing cost = Σ (machine hour rate × cycle time)
                       for every operation completed before rejection
```

Machine hour rate comes from the cost master. **Cycle time per operation is asked once per operation and stored** — the same build-itself mechanic. There is no assumption that a shopfloor data system exists (see Part F); if one does, these values import automatically instead.

Rework works the same way — never store "rework cost per part." Calculate it:

```
Rework cost = rework minutes × (labour rate + machine hour rate, if a machine
is used)
            + any consumables/material added
```

What AI asks per scrap Kaizen — 3 questions, not a rate table

```
AI:  Which part number?
👤   Bracket A-2214

AI:  At which stage is it being rejected?
     [ Blanking ] [ Forming ] [ Welding ] [ Final Insp. ]
👤   Final Inspection

AI:  Roughly how many per month?
👤   ~180
```

That's all. The first time a part appears in any Kaizen, AI asks for its weight, material grade and route **once** — after that it's remembered, and every future Kaizen on that part needs only the three questions above.

Design principle: anything with **hundreds of possible values gets calculated**; only things with **a handful of stable values get stored and asked for.** Applied to scrap, that means asking for 8 material rates once — not maintaining 200 part-level scrap costs that go stale the moment prices move.

⭐ The master builds itself

Every stored answer is **saved permanently**. The MS material rate is asked once, in month 1, and never again. By month 3 a plant has a complete, accurate cost master — assembled entirely from single-field questions asked at the moment they mattered, without anyone ever sitting down to fill in a setup screen.

Three rules so this doesn't become data entry:

1. **Maximum one missing-rate question per Kaizen.** If two are missing, ask for the one with the bigger impact on the number and estimate the other.
2. **Never block approval.** The approver can skip the question and approve anyway — the Kaizen simply stays *Provisional*.
3. **Never ask the operator.** These questions only ever appear on an approver's screen.

Maintenance of the master

- **Owned by one named person** at the plant, reviewed once a year or whenever a rate changes.
- **The bottleneck list is refreshed monthly**, not annually — bottlenecks move with product mix. It's a short manual list the plant maintains (typically 2–5 stations); if a shopfloor data system happens to be live, it can update automatically instead.

⚠️ Two things that will actually happen at a real plant

1. Plants will refuse to share contribution margin. It's commercially sensitive, and a Plant Head will not type their per-part profit into a third-party app in week one. **Don't let this block the pilot.** Fallback ladder, in order:

- Contribution margin per part *(most accurate)*
- → Plant-average contribution as a % of selling price
- → **Machine hour rate as a proxy** *(good enough, and every plant already has this number)*

Design the calculation so it silently drops to the next rung if the number above isn't available.

2. Group B numbers will be missing for the first few weeks — that's by design, not a failure. Until a rate is supplied, the system uses a **clearly labelled default:**

```
₹20,250/yr   ⓘ PROVISIONAL
Using default labour rate ₹150/hr
→ Plant Head to confirm actual rate
```

Any Kaizen calculated on a default is tagged **Provisional** and is **excluded from the verified savings total and the Friday digest** until the rate is confirmed. This prevents the single worst outcome: a Plant Head seeing "₹18 lakh saved" built on numbers the software made up.

Worked example — inputs already collected

From the operator (Step 4)	From the Plant Cost Master
12 trips/shift · 45 sec/trip	Labour rate ₹150/hr
3 shifts	300 working days/yr
₹12,000 proposed cost *(their estimate)*	Press-2 not on bottleneck list

```
Walking time per year
= 45 sec × 12 trips × 3 shifts × 300 days ÷ 3600
= 135 hours per year

Labour cost @ ₹150/hr = ₹20,250 per year
```

> **On the ₹12,000:** that's the operator's guess, and it's fine as a guess — it's only used to rank the idea. It's marked *"operator estimate"* and replaced with the real figure by Maintenance or Purchase at the approval step. Payback and ROI recalculate automatically when the real number lands.

⚠️ **The rule that makes this product credible**

Time saved is only money saved if that time was limiting the plant's output.

If the station is **not** the bottleneck, the operator walks less or the machine cycles faster — but the factory does not ship one extra part, and the wage bill does not change. Calling that

"₹20,250 saved" is how AI tools lose the room in front of a Plant Head.

This applies to time-based Kaizens only — cycle time, changeover, motion, waiting and handling alike. Scrap, rework, energy, consumables and tooling savings go straight into hard saving once verified — money not spent is money saved, whether or not the station is a bottleneck.

So for **time-based** Kaizens, the product **shows three tiers, never one blended figure:**

```
┌──────────────────────────────────────────────┐
│  💰 HARD SAVING (tangible, cash)    ₹0/yr    │
│     This station is not the bottleneck.      │
│     Time saved does not add output today.    │
│                                              │
│  📊 SOFT SAVING (tangible, capacity) ₹20,250 │
│     135 hrs/yr of walking removed            │
│     Real hours — becomes cash when           │
│     redeployed or if this becomes the        │
│     bottleneck                               │
│                                              │
│  🤝 INTANGIBLE BENEFIT           (not costed)│
│     Fatigue ↓ · Safety ↑ · 5S ↑              │
│     → shown as stars, never as rupees        │
└──────────────────────────────────────────────┘
```

A Plant Head who sees an honest ₹0 hard saving will trust the ₹4,00,000 hard saving on the next Kaizen. A Plant Head who catches one inflated number stops believing every number in the system.

Where the bottleneck answer comes from

- **Default (standalone):** the operator's answer in Step 4, plus the supervisor's confirmation at approval. The plant maintains a short bottleneck list, reviewed monthly.
- **If the Kaizen isn't time-based, or isn't tied to a machine or line:** no bottleneck input is needed at all — the saving is calculated directly.
- **Optional accelerator:** if the plant happens to run a shopfloor data system, import the bottleneck flag automatically and skip the question. Never a requirement — see Part F.

STEP 7 — AI CHALLENGES THE IDEA ⭐ *(This is what makes it a consultant, not a form)*

AI does not simply agree and write up whatever was suggested.

Example:

```
┌──────────────────────────────────────────────┐
│  🤔 Before we submit — one alternative       │
│                                              │
│  YOUR IDEA                                   │
│  Buy a second toolbox                        │
│  Investment: ₹12,000                         │
│  Walking removed: 100%                       │
│  Payback: 3.7 months                         │
│                                              │
│  ALTERNATIVE                                 │
│  Move the existing toolbox to the midpoint   │
│  between Machine 2 and Machine 3, on a       │
│  trolley                                     │
│  Investment: ₹800                            │
│  Walking removed: ~92%                       │
│  Payback: 1 week                             │
│                                              │
│  Same result, ₹11,200 less.                  │
│                                              │
│  [ Submit mine ]  [ Submit alternative ]     │
│  [ Submit both — let manager decide ]        │
└──────────────────────────────────────────────┘
```

⚠️ **Four rules so this doesn't backfire**

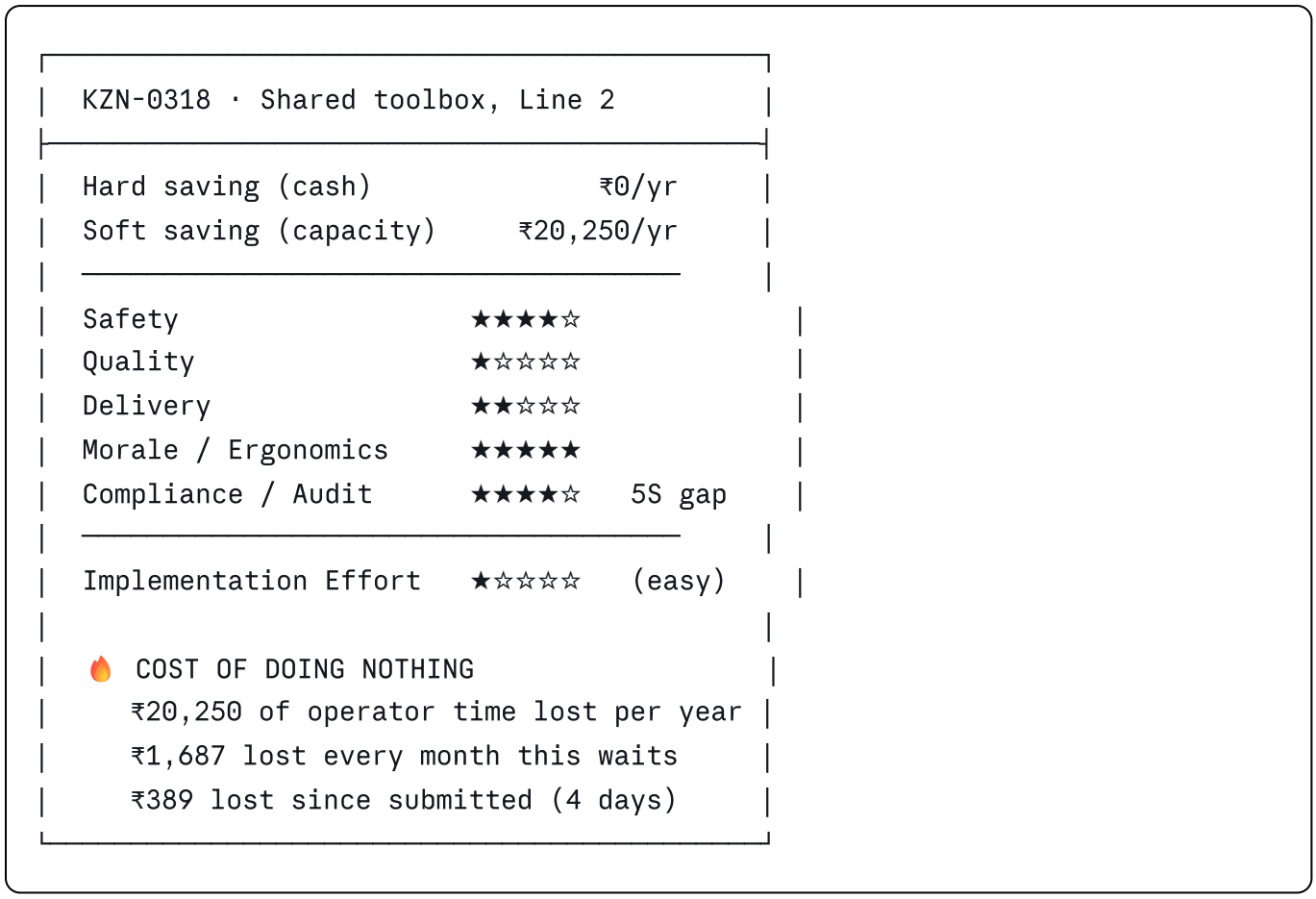
1. **Maximum ONE challenge per Kaizen.** Two feels like an argument.
2. **The operator can always submit their original with one tap.** AI advises; it never blocks or overrules.
3. **Wording is always "an alternative," never "a better idea."** An operator whose idea gets corrected by a machine in front of their supervisor stops submitting. Tone here is a product requirement, not copywriting.
4. **AI does not know the plant layout.** It will sometimes suggest something physically impossible ("move it to the midpoint" — when there's a pillar there). So every challenge ends in a *question*, never an instruction, and every dismissal is logged so the suggestions improve.

Why this feature matters commercially: any competitor can write a Kaizen report with AI. Very few will build something that *argues back with a cheaper option*. This is the demo moment that sells the product in a boardroom.

STEP 8 — The Business Impact Card ⭐ *(Replaces the single score)*

Managers do not approve on ROI alone. A ₹4,000 saving that removes an audit non-conformity gets approved instantly; a ₹90,000 saving that needs a 2-day shutdown sits for six months.

So instead of one number, every Kaizen carries a **Business Impact Card. Money is shown in rupees; everything that can't be costed is shown in stars** — the two never get mixed into a single figure.



On "Cost of Doing Nothing" ⭐

Same number, opposite framing. *"Annual saving ₹20,250"* is a nice-to-have. *"₹1,687 walks out of this factory every month you don't decide"* is a decision.

Add the ticking counter — the amount lost *since the Kaizen was submitted*, growing daily on the manager's screen. Nothing clears an approval backlog faster than a number that goes up while you watch.

Keep a hidden composite score too

The star card is for **humans reading one Kaizen**. The system still needs one number to **sort 400 Kaizens** into a queue. So calculate a 0–100 score behind the scenes — never shown as the headline, only used for ranking and for the "Quick Wins" filter.

Suggested starting weights *(tune after 20+ real Kaizens — don't keep guessing forever)*: Hard savings 30 · Ease 20 · Safety 20 · Quality 15 · Compliance 15.

STEP 9 — AI Predicts What Will Go Wrong ⭐

Shown to the **approver**, not the operator, at the moment of decision:

```
┌──────────────────────────────────────┐
│  📊 IMPLEMENTATION FORECAST          │
│                                      │
│  Probability of completion: 94%      │
│  Expected time: 6 days               │
│                                      │
│  Likely obstacles:                   │
│  ⚠ Purchase approval — HIGH risk     │
│    (avg 11 days for items >₹5,000 here)  │
│  ⚠ Trolley availability — MEDIUM     │
│  ✓ No shutdown required              │
│  ✓ No external vendor needed         │
│                                      │
│  💡 To move faster: pre-approve the ₹800  │
│     spend now — that removes the only  │
│     high-risk step.                  │
└──────────────────────────────────────┘
```

⚠️ The cold-start problem — this must be built in two phases

On day 1 at a new plant there is **zero history**, so there is nothing to predict from. Do not promise a prediction engine that has no data.

- **Phase 1 (day one): rule-based.** Simple, honest logic: *cost > ₹10,000 → purchase-approval risk · needs shutdown → schedule risk · external vendor → lead-time risk.* This alone is useful and works from the first Kaizen.
- **Phase 2 (after ~50 completed Kaizens at that plant): learned.** Now it uses that plant's real history — "purchase approvals here take 11 days on average" — which is far more powerful, and *impossible for a competitor to copy*, because it's built from the customer's own data.

PART B — APPROVAL & IMPLEMENTATION

STEP 10 — Configurable Approval Workflow ⭐

There is no fixed workflow. Every plant is different, and a fixed 7-step chain is exactly how a Kaizen sits untouched for three weeks and operators stop trusting the system.

Setup is **no-code** — the plant picks a template on day one and edits it by dragging:

Template A — Simple *(most small plants)*

```
Operator → Supervisor → Closed
```

Template B — Standard

```
Operator → Supervisor → Plant Head → Maintenance → Closed
```

Template C — Open

```
Operator → Auto-approved → Implemented
```

(Used by plants that want maximum participation and will filter later.)

Plus threshold rules the plant sets itself:

```
IF cost < ₹5,000  AND  score > 80   →  auto-approve
IF safety-flagged                    →  always route to Plant Head
IF cost > ₹25,000                    →  add Finance
```

⚠️ **Build 3 templates + a simple editor — not a full workflow builder**

"Fully configurable" sounds good in a sales pitch and becomes a six-month engineering project with a drag-and-drop rules engine nobody uses correctly. **Ship 3 presets, an editable step list, and threshold rules.** That covers ~95% of real plants. Add complexity only when a paying customer is blocked without it.

STEP 11 — AI Auto-Creates the PDCA Project ⭐

The moment a Kaizen is approved, **nobody has to create a project.** AI does it.

Example — auto-generated the instant KZN-0318 is approved:

```
┌──────────────────────────────────────────────────┐
│  PDCA · KZN-0318 — Relocate shared toolbox       │
│  Owner: Maintenance Supervisor                   │
├──────────────────────────────────────────────────┤
│  PLAN                                  Due 28/07 │
│  ☐ Confirm midpoint location            (Maint.) │
│  ☐ Raise PR for trolley ₹800            (Maint.) │
│                                                  │
│  DO                                    Due 02/08 │
│  ☐ Procure trolley                    (Purchase) │
│  ☐ Relocate toolbox + label positions   (Maint.) │
│  ☐ Brief all 4 operators               (Supervisor)│
│                                                  │
│  CHECK                                 Due 16/08 │
│  ☐ Re-time the walk (target <5 sec)  (Supervisor)│
│  ☐ Confirm no obstruction to aisle     (Safety)  │
│                                                  │
│  ACT                                   Due 20/08 │
│  ☐ Update standard work instruction  (Industrial│
│  ☐ Check same fix on Lines 1, 3, 4      Engineer)│
├──────────────────────────────────────────────────┤
│  🔔 WhatsApp reminders auto-sent on due dates    │
└──────────────────────────────────────────────────┘
```

No project manager. No separate tracker. No one typing tasks into Excel.

The final "ACT" step — *check whether the same fix applies elsewhere in the plant* — is deliberately included every time. That's how one Kaizen becomes four.

STEP 12 — Implementation Tracker

Simple status chips the owner taps: `Started` → `50%` → `Waiting for Purchase` → `Shutdown Needed` → `Completed`

Every status change auto-notifies the **original operator** on WhatsApp:

> *"Your Kaizen KZN-0318: trolley ordered, arriving in 5 days."*

Why this small thing matters more than it looks: the single most common reason suggestion schemes collapse is operators believing their idea vanished into a drawer. One WhatsApp message per status change is the cheapest trust mechanism in the entire product.

STEP 13 — Closure & Savings Verification

Two weeks after implementation (a sustain period — not the day after, when everything always looks perfect):

```
AI: Your Kaizen was implemented 14 days ago.
    Did it work?

    📷 Take an AFTER photo
    ⏱ Time the new walk once — we'll compare
    Did the improvement hold?
    [ Yes ] [ Partly ] [ No — it slipped back ]
```

AI compares before/after photos with Vision AI, compares the timing to the original claim, then marks:

```
✅ VERIFIED — Actual saving ₹18,900/yr
   (claimed ₹20,250 · achieved 93%)
```

Only verified savings count towards the plant's total. This is what allows the Plant Head to state a savings figure to their MD, or an auditor, without it being self-reported guesswork.

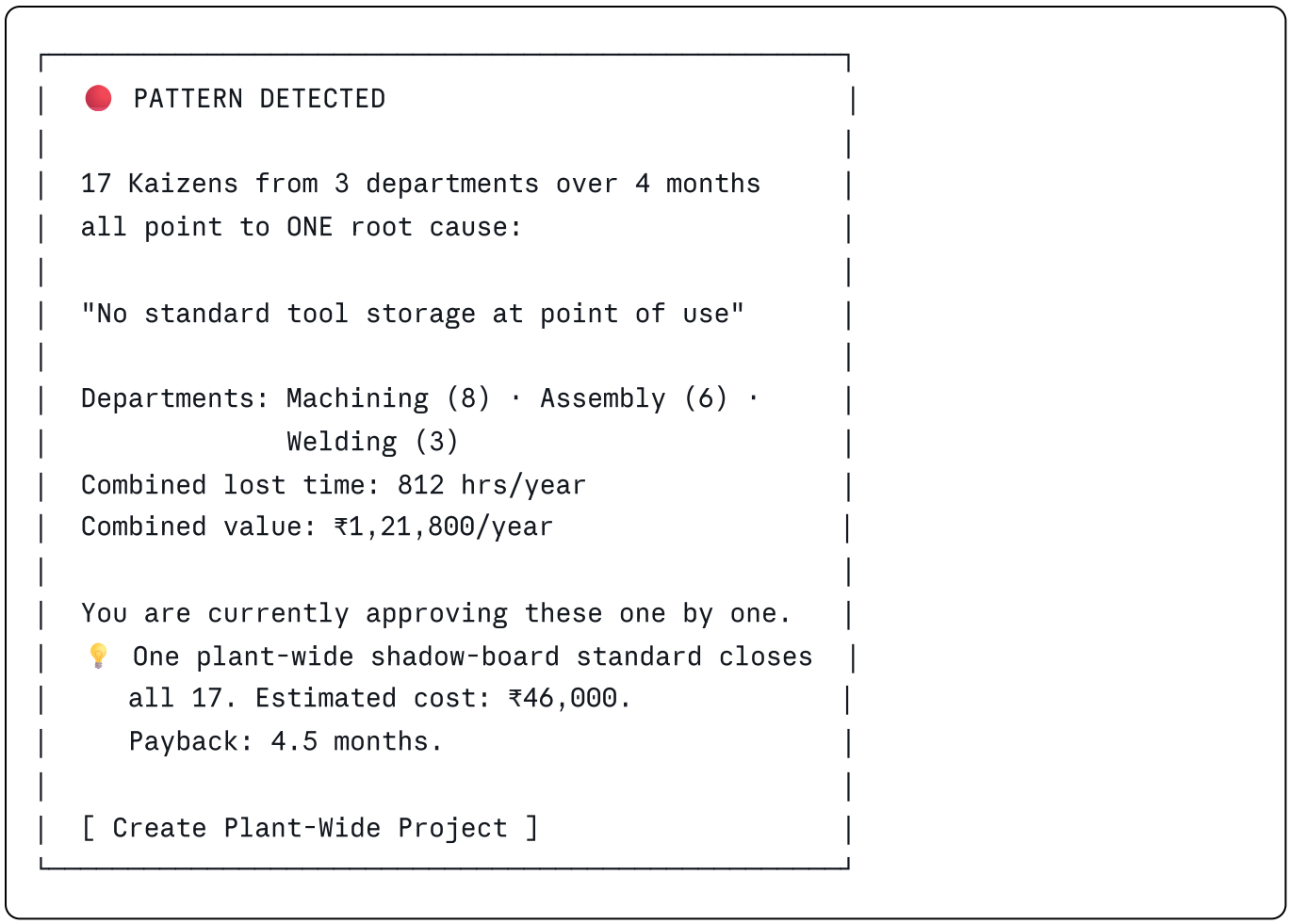
PART C — THE MANAGEMENT LAYER

(Things that happen with nobody clicking anything)

FEATURE 1 — The Repeat Problem Detector ⭐

AI continuously compares every Kaizen in the plant and finds the **hidden common root cause.**

Example — alert that appears on the Plant Head's screen:



This is the highest-value feature in the product, and it is impossible without the plant's own accumulated history — which means it gets *better the longer they stay*, and a competitor starting fresh cannot replicate it. Nobody in the plant can see this pattern manually, because the 17 Kaizens sit in three different departments, across four months, described in seventeen different ways.

How it works underneath: every Kaizen is converted to a mathematical "meaning fingerprint" when submitted; a weekly job groups similar fingerprints together and alerts when a cluster passes a threshold (e.g. 5+ related items, or ₹50,000+ combined value).

FEATURE 2 — The Friday Executive Digest ⭐

(The feature that sells the product)

Every Friday, 7:00 AM. **One page. On WhatsApp and email. No login. No dashboard. No training.**

```
═══════════════════════════════════════
  KAIZEN WEEKLY · Plant: Unit-1 · Wk 30
═══════════════════════════════════════

  THIS WEEK
  New submitted        34    ↑ 12%
  Approved             21
  Implemented          12
  Waiting your action   6    ▲
  Delayed >14 days      3    🔴

  MONEY
  Verified savings this month   ₹2,84,000
  Verified savings YTD          ₹18,60,000
  Pending approval value        ₹4,12,000
  🔥 Cost of doing nothing      ₹34,300/mo

  🏆 BIGGEST OPPORTUNITY THIS WEEK
  KZN-0331 · Die change time reduction
  ₹4,80,000/yr · Payback 2 months
  → Needs your approval

  🔴 TOP RECURRING PROBLEM
  17 Kaizens = one root cause:
  no point-of-use tool storage
  → One project closes all 17

  👤 TOP SHOPFLOOR:  R. Kumar, Line 2 (9)
  👤 TOP SUPPORT:    S. Devi, Quality (6)
  📊 DEPARTMENTS (participation %)
     1.Machining 78%  2.Assembly 64%
     3.Welding 51%    4.Paint 40%

  ✅ DO THESE 3 THINGS NEXT WEEK
  1. Approve KZN-0331 (₹4.8L waiting)
  2. Clear 6 pending approvals (avg 9 days)
  3. Decide on plant-wide tool storage
═══════════════════════════════════════
```

Why this, and not the dashboard, is the sales feature: a Plant Head will not log into a dashboard. They will read one page on their phone on Friday morning. This digest is the *only* part of the product a senior person needs to touch — and it's what makes them approve the renewal, because it puts a rupee figure and their own name next to it every single week.

Build this early. It is not a reporting feature; it is the retention mechanism.

FEATURE 3 — Management Dashboard

For the Lean/IE person who *does* log in daily:

View	What it shows
⚡ **Quick Wins**	High score, low cost, no shutdown — *build this filter first*
Pending Approval	Sorted by cost-of-doing-nothing, ticking
Delayed	Anything >14 days in one status
Top Savings	Verified only
Safety Kaizens	Separate view — safety never competes with ROI for attention
Department Ranking	Participation % , not just count
Employee Ranking	Two separate boards — Shopfloor and Support/Staff (see Part F)
Recurring Problems	Output of the pattern detector

Quick Wins is the view that gets used in week 1. The rest is reporting, and reporting has no value until there's volume. Build Quick Wins first, the others later.

FEATURE 4 — Levels & Gamification

Progression beats a points number:

```
OPERATOR → IMPROVER → LEAN CHAMPION → PROBLEM SOLVER
        → INNOVATION LEADER → SENSEI
```

Each level unlocks a visible badge on the operator's profile, the shopfloor display board, and the Friday digest.

⚠️ Points must be earned on OUTCOME, not on submission count

This is the rule that protects the entire product:

Action	Points
Kaizen submitted	5
Approved	15
Implemented	30
Savings VERIFIED after 2 weeks	**50 + bonus by value**
Safety Kaizen implemented	+25
Idea reused by another department	+40
Duplicate of an existing Kaizen	0

If you award points for submitting, you will get 500 junk submissions a month — every one of which costs money to process, clogs the approval queue, and buries the good ideas. Weighting points towards *verified* outcomes means the operator chasing "Sensei" is chasing exactly what the Plant Head wants.

The "idea reused by another department" bonus is deliberate: it makes operators actively spread their own fixes across the plant.

PART D — THE VOICE ROADMAP

Voice is built in **three phases**, not all at once.

Phase 1 — Voice-to-text only *(launch)*

- Operator **speaks the problem** in Step 1 → converted to text.
- Handles mixed language (regional language + English, the way people actually talk on a shop floor).
- The AI Interview (Step 4) stays **tap-and-type**: buttons for ranges, numbers typed.

> **Why not full voice on day one:** free-form description transcribes well. Short numeric answers in a noisy plant do not — "fifteen" and "fifty" are one machine hum apart. A wrong number here silently corrupts the savings calculation, and a wrong savings number is the one thing that kills credibility with a Plant Head. Get the money right first; make it more convenient second.

Phase 2 — Voice-driven interview *(after ~3 months of real plant data)*

- AI **reads each question aloud** in the local language; operator answers by voice.
- **Every number is shown on screen for a confirmation tap** before it enters the calculation. Never a silently voice-parsed number.

Phase 3 — Fully conversational *(later)*

- Free back-and-forth, like exchanging WhatsApp voice notes.

- Multi-language, auto-detected.
- AI probes using plant history: *"Someone on Line 1 raised a similar toolbox issue in March — is this the same root cause?"* — which requires the plant memory from Step 5 and the pattern detector to already be live.

PART E — WHAT WE ARE DELIBERATELY *NOT* BUILDING IN V1

Saying no is what keeps the 3-minute promise:

Not building	Why
Hardware / IoT sensors	Software-only. Hardware kills speed of deployment.
Full drag-and-drop workflow builder	3 templates + threshold rules covers ~95% of plants
Voice-answered numbers	Accuracy risk to the savings engine (Phase 2)
Full learned prediction engine	No data on day one — rule-based first
Cross-plant benchmarking	Needs 40–50 live plants before the numbers mean anything
A desktop app for operators	Operators have phones, not desks

PART F — LOCKED DECISIONS

✅ DECIDED — Standalone product, first and always

Kaizen Manager AI is a standalone product. It works in any plant — small, medium or large — with **no dependency on any other system.** No shopfloor data system, no MES, no ERP, no hardware.

Why this is the right call: it can be sold to any factory from day one, piloted in a week, and it doesn't wait on anything else being live. That's a market of every manufacturing plant, not the small subset that already has a data system.

What being standalone means for the build — 5 consequences

Consequence	How it's handled
1. No automatic bottleneck flag	Asked in Step 4 on time-based Kaizens, confirmed by the approver. Plant maintains a simple bottleneck list, reviewed monthly.
2. No cycle time data available	Cycle times are asked once per operation, on demand, and stored — same "master builds itself" mechanic as the cost rates.
3. No part master to read from	A lightweight part record builds itself: the first time a part appears in a Kaizen, AI asks weight, material grade and route once, then remembers it forever.
4. No live production data to verify savings against	Verification in Step 13 is done by re-timing and re-counting on the floor, plus before/after photos — which is what a Lean auditor does anyway, and is defensible in front of a customer audit.
5. Cost master is fully manual	Group A at enrollment, Group B on demand. Already designed for this.

Integration is optional, and comes later

If a plant *does* have a shopfloor data system, ERP or MES, offer a **read-only import** for bottleneck status, cycle times and part masters — as an accelerator that removes the manual questions, never as a requirement. **No integration work in v1.** A product that needs an integration before it can be demonstrated cannot be sold in a first meeting.

Selling to large plants

Large plants usually already have an ERP and often a suggestion system on paper. The pitch there is not "we store your ideas" — it's the four things their existing system cannot do: **AI writes the report, AI challenges the idea, AI finds the repeat root cause across departments, and the Plant Head gets one page on Friday.** Standalone means you can prove all four in a 45-minute demo without their IT department being in the room.

✅ DECIDED — Volume target: 10 submissions per employee per month

Volume scales with headcount, not with plant turnover. The design target is **~10 submissions per employee per month** in a small plant, which reconciles to the 200–500/month figure a small plant is expected to generate.

Plant size	Headcount	Target/month	Target/year
Small	30–50	300–500	3,600–6,000
Medium	100–250	800–2,000	10,000–24,000
Large	500+	3,000+	36,000+

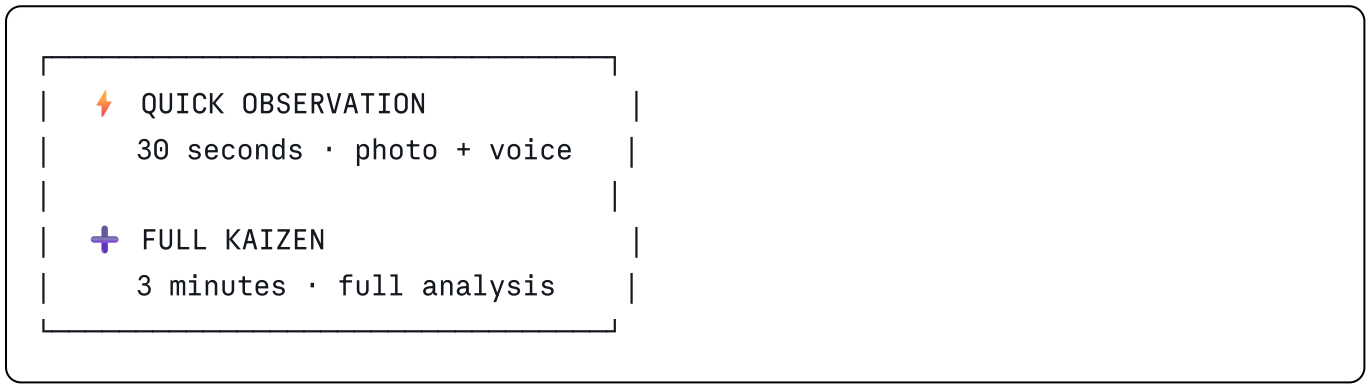
(Per-employee rate typically falls as headcount rises — a 500-person plant will not sustain 10/head/month. Treat medium and large as ranges to confirm from the first pilot at that size.)

⚠️ **At this volume, one design change becomes mandatory: TWO TRACKS**

10 per employee per month is **one idea every two working days per person.** At that rate, most submissions will be small observations — a loose guard, an unlabelled bin, a leaking air line — not engineered improvements.

Running the full AI interview, cost engine, challenge and impact card on a ₹200 observation is wasted processing and, worse, **it buries the four ideas a month that are actually worth ₹4 lakh.**

So the home screen offers **two buttons, not one:**



	⚡ Quick Observation	➕ Full Kaizen
Operator time	**30 seconds**	3 minutes
Steps	Photo + voice only	Full journey, Steps 1–9
AI interview	None	Yes, 4–6 questions
Cost engine	None	Yes
AI Challenge	None	Yes
Approval	Auto-routed to the owning department, no approval needed	Full workflow
Points	5	5 → 50 by outcome
Expected share of volume	**~80%**	~20%

AI auto-promotes when it matters. If a Quick Observation looks significant — recurring, safety-related, or estimated above a value threshold — AI comes back to the operator:

> *"This one looks worth a proper analysis. 4 quick questions?"*

Why this protects everything else in the product: it keeps the approval queue survivable (a supervisor with 40 operators facing 400 items/month cannot review 400), it keeps processing cost proportional to value, and it keeps the Friday digest showing the ideas that matter instead of a list of 400 housekeeping notes.

✅ DECIDED — Operators are the primary user; everyone else is welcome

Anyone in the plant can submit — operators, supervisors, maintenance, quality, stores, office, dispatch. But the product is **designed for the operator first**: the phone screen, the voice input, the 30-second Quick Observation and the 3-minute Full Kaizen all exist because an operator has neither a desk nor time.

What this settles: the home screen stays as designed — **single entry by many people**, not bulk entry by one Lean engineer. No bulk-entry mode is built.

Three things this makes necessary

1. Submit-on-behalf, with credit to the operator A supervisor will sometimes submit for an operator who is uncomfortable with the phone — especially in month one. So the submit screen has one optional field:

```
Submitting on behalf of:  [ R. Kumar, Line 2 ▾ ]
```

Points and recognition go to the named operator, not the submitter. Get this wrong and the first time an operator sees their idea credited to their supervisor, participation collapses on that line and never recovers.

2. Nobody approves their own Kaizen When a supervisor or manager submits, the Kaizen automatically routes **one level up**. Otherwise the approval workflow is theatre, the savings numbers are self-certified, and the whole audit trail is worthless in front of a customer.

3. Separate the leaderboards — this is the one that will quietly break the product If a quality engineer competes with a machine operator on the same ranking, the engineer wins every month. They have better vocabulary, more context, more exposure to the whole plant and more time at a desk. The operator sees a leaderboard they can never top and stops submitting.

So run **two rankings, always shown side by side:**

Ranking	Who competes	Measured on
Shopfloor	Operators, technicians	Points from verified outcomes
Support & Staff	Supervisors, engineers, office	Points from verified outcomes

And rank departments on **participation** % (how many of that department's people submitted at least one this month), not raw submission count — otherwise a 60-person machine shop always beats a 4-person stores team, and stores stops trying.

Non-production departments are fully supported — stores, quality, office and dispatch Kaizens simply skip the bottleneck question and the machine-hour calculations, exactly as designed in Step 4.

✅ DECIDED — Three tiers, mapped onto what plants already use

Plants today classify Kaizen benefits as **tangible** and **intangible**. That vocabulary stays — the product does not invent new language. What it adds is one split *inside* tangible, and that split is the credibility engine.

Tier	Plant's existing word	Question it answers	Shown as
💰 **Hard saving**	Tangible	Does this hit the P&L this year?	₹ — counts in the plant total
📊 **Soft saving**	Tangible	Real hours or units recovered, but not cash yet	₹ — reported separately, never added to the plant total
🤝 **Intangible benefit**	Intangible	Can't be costed at all	★ stars on the Business Impact Card

Worked examples:

Kaizen	Hard	Soft	Intangible
Scrap reduction, 180 parts/mo	₹6,26,400	—	Customer confidence ★★★★☆
Cycle time −6 sec at bottleneck press	₹8,10,000	—	—
Cycle time −6 sec at a non-bottleneck machine	₹0	₹1,52,000 capacity	—
Toolbox relocation, 135 hrs walking	₹0	₹20,250	Fatigue ★★★★★ Safety ★★★★☆
Guard fitted on a pinch point	₹0	—	Safety ★★★★★
Air leak fixed, 4 CFM	₹31,000	—	Energy/ESG ★★★☆☆

Why this is better than the single "savings" number plants use today, and how to sell it:

- The MD gets a number they can take to the board **and defend under audit**, because every rupee in it is traceable to cash.
- Nothing is lost — soft savings and intangibles are still reported, still visible, still earn the operator points. They're just in the right column.

- A plant that has been reporting one blended figure has, almost always, been **overstating it.** The first year on this system may show a *lower* headline number than last year's. That's the sales conversation to prepare for, and the honest answer is: last year's number was never defensible; this one is.

Hold this line even when a plant asks for one merged figure. Some will ask. The two-column split is what makes the Friday digest credible in front of an auditor or a customer, and once you merge it for one plant you have no product differentiator left — every competitor can print one big number.

✅ DECIDED — Minimum one submission per employee per month

The plant sets a floor: **every employee submits at least one per month.** The software measures and reports it; the plant owns the policy. This is already common practice in TPM and IATF plants, so it needs no selling — it needs supporting.

How it works:

Minimum	1 per employee per month *(plant-configurable: 0, 1, 2…)*
Satisfied by	**Either track** — a 30-second Quick Observation counts
Measured as	Participation % per department, already on the Friday digest
Employee sees	`This month: 0 of 1 · 6 days left` on their home screen
Supervisor sees	A list of who hasn't submitted yet — a **nudge list, not a punishment list**

⚠️ Every mandatory quota in industrial history has produced junk on the last day of the month

This is the one real risk of the rule, and it works against three things already built into this product: processing cost per submission, a survivable approval queue, and not burying the four ideas a month worth ₹4 lakh under four hundred filler notes.

Four rules that contain it:

1. **The floor is deliberately low and easy.** One 30-second Quick Observation clears it. Nobody needs to invent a fake improvement to comply — noticing one real thing in a month is a genuinely low bar. Do not raise the default above 1.
2. **Quantity clears the floor; only quality earns points.** The mandate and the leaderboard are separate systems. Submitting one filler observation satisfies the rule and earns 5 points. The operator chasing *Sensei* is still chasing verified savings — see the points table in Part C.
3. **Duplicates count for nothing.** A near-identical repeat of something already in the plant's history scores 0 points **and does not clear the month's requirement.** AI already detects this in Step 5; it just has to be wired to the counter.

4. **AI flags end-of-month spikes to the Plant Head.** If a department submits 80% of its month's volume in the last two days, the Friday digest says so plainly. The point isn't to punish anyone — it's to tell the Plant Head their scheme has become a tick-box exercise on that line, which is exactly what they'd otherwise find out a year too late.

And one honest line for the sales conversation: participation % is a leading indicator, savings is the lagging one. A plant at 95% participation and ₹40,000 of verified savings has a compliance culture, not an improvement culture. Report both, always together, and never let the participation number become the headline in the Friday digest.

THE 30-SECOND VERSION FOR ANYONE

> An operator talks to their phone for 30 seconds and takes one photo.
> AI reads the photo and spots things they didn't mention. It asks 5 quick questions. It writes the full Kaizen report. It calculates honest money — separating cash savings from comfort savings. It suggests a cheaper alternative. It checks whether 16 other people already reported the same thing. It predicts what will delay the fix.
> The manager sees a star-rated impact card and a number showing what this costs the factory every month it isn't decided. They tap approve, and the entire PDCA project creates itself with owners, dates and reminders.
> Two weeks after it's done, AI verifies whether the savings actually happened.
> And every Friday morning the Plant Head gets one page telling them exactly what to do next week.
> **Operator effort: 3 minutes. Manager effort: one tap. Everything else is AI.**